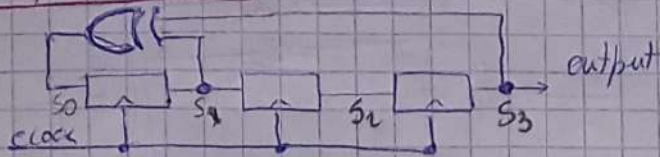


Aula Teórica 9 (21/11/2023)

Pseudo Noise Sequence

Exemplo (Slide 8)



State	Output
1 0 0	1
1 1 0	0
1 1 1	1
0 1 1	1
1 0 1	1
0 1 0	0
0 0 1	1
1 0 0	0

volta-se a repetir ao fim de 7 estados, ou seja, o período é igual a $2^n - 1$, $n=3$ (n° flip-flops)

$$\text{output} = \overbrace{0011101}^{\text{período}} \dots$$

	S_0	S_1	S_2	S_3
A	1	1	0	0
B	1	1	1	0
C	0	1	1	1
D	1	0	1	1
E	0	1	0	1
F	0	1	0	1
G	0	0	1	0
H	1	0	0	1
A	1	1	0	0
B	1	1	1	0
C	0	1	1	1
D	1	0	1	1
E	0	1	0	1

período

Aula Teórica 10 (28/11/2023)

Exercício (Slide 12)

$$s_1(t) = \cos(\omega t + \phi_1)$$

$$s_2(t) = \cos(\omega t + \phi_2)$$

$$s_1(t) + s_2(t) = \cos(\omega t + \phi_1) + \cos(\omega t + \phi_2) \quad \left(\cos \alpha + \cos \beta = 2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2} \right)$$

$$= 2 \cos \left(\frac{2\omega t + \phi_1 + \phi_2}{2} \right) \cdot \cos \left(\frac{\phi_1 - \phi_2}{2} \right)$$

$$= 2 \cos \left(\frac{\phi_1 - \phi_2}{2} \right) \cos \left(\omega t + \frac{\phi_1 + \phi_2}{2} \right)$$

$$\text{Const.} = A, \text{ se } \phi_1 - \phi_2 = 180 \Rightarrow \frac{\phi_1 - \phi_2}{2} = 90$$

$$\Rightarrow \cos \left(\frac{\phi_1 - \phi_2}{2} \right) = 0$$

Exercício (Slide 34)

$$f_D = \pm f_0 \frac{v}{c}$$

$$f_0 = 900 \text{ MHz}, \quad v = 120 \text{ Km/h}$$

$$= \pm 900 \times 10^6 \times \frac{120}{1080 \times 10^3} = \pm 100 \text{ Hz}$$

Exemplo (Slide 64)

$$B_{\text{OFDM}} = f_0 + (K-1)\Delta f + 2\Delta f$$

$$= f_0 + K\Delta f - \Delta f + 2\Delta f$$

$$= f_0 + K\Delta f + \Delta f$$

$$= f_0 + (K+1)\Delta f$$

Exercício (Slide 68)

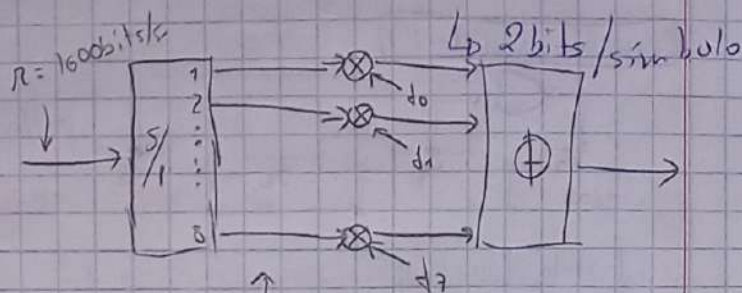
$$R = 1600 \text{ bits/s}$$

OFDM

$$K = 8$$

$$QPSK \Rightarrow M = 4 \Rightarrow n = \log_2(M) = 2$$

$$f_0 = 650 \text{ Hz}$$



$$a) f_7 = ?$$

$$f_7 = f_0 + (K-1) \times \Delta f$$

$$= 650 \text{ Hz} + 7 \times 100$$

$$= 1350 \text{ Hz}$$

$$R_b = \frac{1600}{8} = 200 \text{ bits/s}$$

$$R_A = \frac{R_b}{2} = \frac{200}{2} = 100 \text{ bits/s}$$

$$\Delta f = R_A = 100 \text{ bits/s}$$

$$b) B_{\text{OFDM}} = (K+1) \times \Delta f$$

$$= 9 \times 100$$

$$= 900 \text{ Hz}$$

$$c) B_{\text{SC}} = \frac{R_b \times 2}{n} = R_b = 1600 \text{ Hz}$$

$$a) K = 8 \text{ portadoras}, \underbrace{f_0, \dots, f_7}_{\text{8 portadoras}} \rightarrow ?$$

$\parallel$
 650 Hz

$$f_\alpha = f_0 + \alpha \Delta f \leftarrow \text{formula geral para qualquer subportadora}$$

$$\alpha = 7$$

$$f_0 + 7 \times \Delta f = 650 + 7 \times 100 = 1350 \text{ Hz}$$

$$b) B_{\text{OFDM}} = (K+1) \times \Delta f$$

$$= (8+1) \times \Delta f$$

$$= 9 \times 100 = 900 \text{ Hz}$$

$$c) B_{\text{SC}} = \frac{R \times 2}{n}$$

$$= \frac{1600 \times 2}{2}$$

$$= 1600 \text{ Hz}$$

Aula Teórica 11 (12/12/2023)

$$x[n] \longleftrightarrow X[k]$$

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad 0 \leq k \leq N-1 \implies \text{Análise (Eq.)}$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}, \quad 0 \leq k \leq N-1 \implies \text{Síntese (Eq.)}$$

$$W_N = e^{-j\left(\frac{2\pi}{N}\right)}$$

Exemplo Slide 12

$$\tilde{X}[k] = \sum_{n=0}^4 e^{-j\left(\frac{2\pi k}{5}\right)n}$$

$$= e^{-j\left(\frac{2\pi k}{5}\right) \times 0} + e^{-j\left(\frac{2\pi k}{5}\right) \times 1} + e^{-j\left(\frac{2\pi k}{5}\right) \times 2} + e^{-j\left(\frac{2\pi k}{5}\right) \times 3} + e^{-j\left(\frac{2\pi k}{5}\right) \times 4}$$

$$= 1 + e^{-j\left(\frac{2\pi k}{5}\right) \times 1} + \dots + e^{-j\left(\frac{2\pi k}{5}\right) \times 4}$$

Prog. Geo.

$$a_0 = 1$$

$$q = e^{-j\left(\frac{2\pi k}{5}\right)}$$

$$\implies S_5 = \frac{1 - e^{-j(2\pi k)}}{1 - e^{-j\left(\frac{2\pi k}{5}\right)}} = \tilde{X}[k] \quad S_N = \frac{1 - q^N}{1 - q}$$

$$\tilde{X}[1] = \frac{1 - e^{-j(2\pi)}}{1 - e^{-j\left(\frac{2\pi}{5}\right)}} = \frac{1 - 1}{1 - \dots} = \frac{0}{1 - \dots} = 0 \quad e^{-j(2\pi)} = \cos(2\pi) - j\sin(2\pi) = 1$$

$$\tilde{X}[5] = \frac{1 - e^{-j(2\pi \cdot 5)}}{1 - e^{-j(2\pi)}} = \frac{1 - 1}{1 - 1} = \frac{0}{0}$$

$$\text{Ind: } \tilde{X}[k] = \frac{(1 - e^{-j(2\pi k)})}{(1 - e^{-j\left(\frac{2\pi k}{5}\right)})} = \frac{+j2\pi \cdot e^{-j2\pi k}}{+j\frac{2\pi}{5} \cdot e^{-j\frac{2\pi k}{5}}}$$

$$k=0 \implies \tilde{X}[0] = \frac{j2\pi \cdot 1}{j\frac{2\pi}{5} \cdot 1} = \frac{1}{1/5} = 5$$

DTFT

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] \cdot e^{-j\omega n}$$

↳ função complexa periódica ($T=2\pi$) contínua

$$x_k(t) = I_k(t) \cdot \cos(2\pi f_p t) - j Q_k(t) \cdot \sin(2\pi f_p t)$$

Slide 15

$$\begin{aligned} x_k(t) \cdot e^{j2\pi f_k t} &= I_k \cdot \cos(2\pi f_p t) \cdot e^{j2\pi f_k t} - j Q_k(t) \cdot \sin(2\pi f_p t) \cdot e^{j2\pi f_k t} \\ &= I_k \cos(2\pi f_p t) \cdot (\cos(2\pi f_k t) - j \sin(2\pi f_k t)) - j Q_k(t) \cdot \sin(2\pi f_p t) \cdot (\cos(2\pi f_k t) - j \sin(2\pi f_k t)) \end{aligned}$$

Slide 22

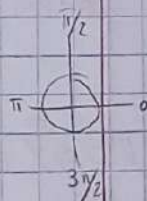
$$X_k(t) = I_k(t) + j Q_k(t)$$

$$\text{Data} = [010 \ 011 \ 010 \ 110]$$



$$W(n) = \frac{1}{K} \sum_{k=0}^{K-1} X_k \cdot e^{j \frac{2\pi n k}{K}}$$

$$\begin{aligned} W(0) &= \frac{1}{4} \left(\sum_{k=0}^3 X_k \cdot e^0 \right) = \frac{1}{4} (X_0 + X_1 + X_2 + X_3) \\ &= \frac{1}{4} (1+j + (-1+j) + (-1-j) + (-1-j)) \\ &= \frac{1}{4} (-2 + 2j) \\ &= -0,5 + 0,5j \end{aligned}$$



$$\begin{aligned} W(1) &= \frac{1}{4} \left(\sum_{k=0}^3 X_k \cdot e^{j \frac{2\pi k}{4}} \right) = \frac{1}{4} \left(X_0 \cdot e^0 + X_1 \cdot e^{j\frac{\pi}{2}} + X_2 \cdot e^{j\pi} + X_3 \cdot e^{j\frac{3\pi}{2}} \right) \\ &= \frac{1}{4} \left((1+j) \cdot 1 + (-1+j) \cdot (\cos(\frac{\pi}{2}) + j \sin(\frac{\pi}{2})) + (-1-j) \cdot (\cos(\pi) + j \sin(\pi)) + (-1+j) \cdot (\cos(\frac{3\pi}{2}) + j \sin(\frac{3\pi}{2})) \right) \\ &= \frac{1}{4} \left((1+j) \cdot 1 + (-1+j) \cdot j + (-1-j) \cdot (-1) + (-1+j) \cdot (-j) \right) \\ &= \frac{1}{4} \left((1+j) + (-j+j^2) + (1+j) + (j-j^2) \right) \\ &= \frac{1}{4} \left((1+j) + (-j-1) + (1+j) + (j+1) \right) \\ &= \frac{1}{4} \left((1+j) + (-1-j) + (1+j) + (1+j) \right) \\ &= \frac{1}{4} \left((1+j) + (1+j) \right) \\ &= \frac{1}{4} (2(1+j)) \\ &= \frac{1}{2} (1+j) \\ &= 0,5 + 0,5j \end{aligned}$$

Euler

$$e^{j\theta} = \cos(\theta) + j \sin(\theta)$$